

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1501
COURSE OUTCOME:	CO4: Analyze big data characteristics and challenges across domains including security and application aspects. (BL4)		
WEIGHTAGE / MARKS:	Weightage: 40% Total Marks: 40	TAXONOMY / DURATION:	Dave's Level 4 (Articulation) Duration: 60 Mins Q. Count: 5

CODE OF CONDUCT

I Aakash R (192524462) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Aakash R

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Q1 - Retail Data Optimization

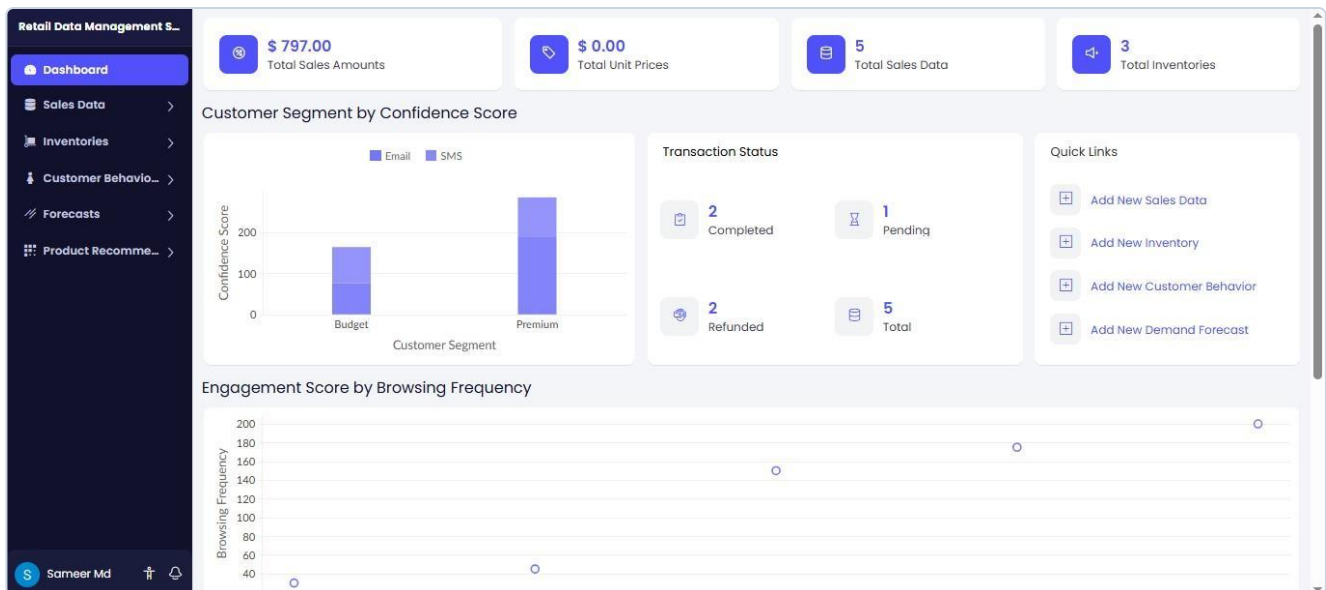
Problem Statement: A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

A retail company generates large volumes of sales, inventory, and customer behavior data from physical stores and its online platform. A scalable Big Data architecture can integrate these different data sources and process them in real time. The proposed architecture collects data from POS systems, e-commerce transactions, inventory systems, customer interactions, and website/mobile applications. The data can be stored in a scalable cloud data lake or distributed storage system. **Apache Kafka** can be used for real-time data streaming, while **Apache Spark / PySpark** can process both real-time and historical data.

- Real-time inventory tracking:** Streaming data from physical stores and online orders is processed continuously to update stock levels dynamically, preventing stock-outs or costly overstocking.
- Demand forecasting:** Historical sales data is analyzed using machine learning models such as Linear Regression, Random Forest, or ARIMA/Prophet time-series forecasting to forecast seasonal and localized demand.
- Personalized marketing:** Customer interactions (purchases, searches, browsing duration) are analyzed using collaborative filtering and customer segmentation techniques to generate personalized product recommendations.



Scalability & Security: Distributed Spark/Kafka processing ensures horizontal scalability as transaction volumes grow. Security is enforced via TLS encryption, Kerberos authentication, RBAC, and automated audit logging.



All Sales Data

Transaction Id	Product Name
T005	Smart Watch
T004	Yoga Mat
T003	Running Shoes
T002	Organic Coffee Beans
T001	Wireless Headphones

Overview of Transaction T005:

Transaction Id	T005
Product Name	Smart Watch
Category	Electronics
Channel	Online
Quantity Sold	2
Sales Amount	\$ 350.00
Transaction Date	18-Aug-2026 16:00:00
Product Reference	
Store Location	City Center
Customer Segment	Premium
Transaction Status	Refunded

Product Reference: No records found

Q2 – Social Media Fraud Detection

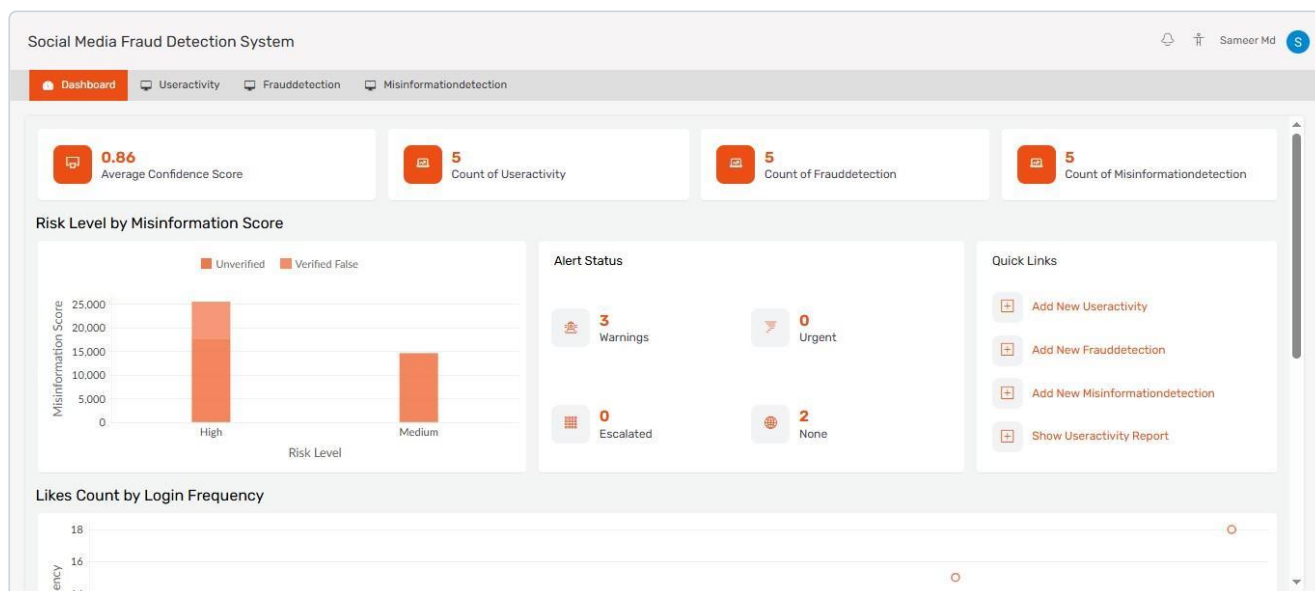
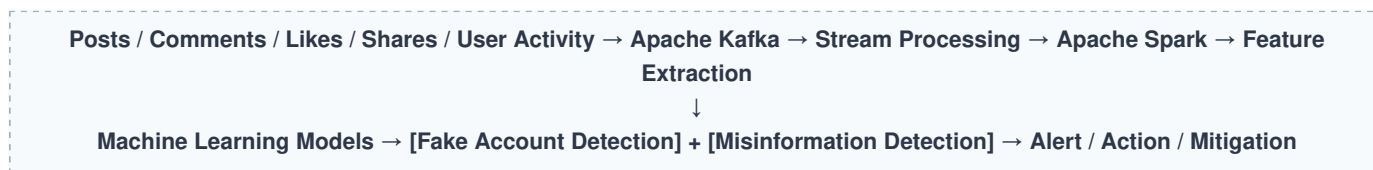
Problem Statement: A social media platform processes billions of user interactions daily (posts, comments, likes, shares). The platform wants to detect fake accounts and misinformation quickly in real-time or near real-time. Design a scalable big data solution and explain the role of machine learning and stream processing frameworks.

A scalable Big Data fraud detection system collects high-velocity event streams from posts, comments, likes, shares, user profiles, login activities, and interaction patterns. A stream-processing architecture enables near-real-time behavioral and content analysis.

Apache Kafka serves as the central streaming backbone, ingesting event streams from web and mobile apps. **Apache Spark Streaming / PySpark** extracts real-time features including posting velocity, repeated content hashing, unusual geo-location jumps, and bot-like interaction bursts.

- **Fake Account Detection:** Supervised classification models (Random Forest, Logistic Regression, XGBoost) classify accounts based on behavioural metrics and network topology.

- **Misinformation Detection:** Natural Language Processing (NLP) models and sentiment/fact-checking pipelines examine textual post content and propagate credibility scores.
- **Feedback Loop:** Historical data in cloud data lakes periodically retrains ML models, while real-time inference engines trigger instant account throttling, warnings, or auto-blocks.



Social Media Fraud Detection System

Dashboard | **Useractivity** | Frauddetection | Misinformationdetection

Useractivity Report

Username	Activity Type	Posts Count	Comments Count	Likes Count
eve_writer_33	Login	95	112	430
diana_gamer_77	Post	310	220	110
charlie_dev_x	Post	201	156	89
bob_trader_01	Comment	87	45	21
alice_fan_99	Share	142	89	52

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Overview

Username	eve_writer_33
Activity Type	Login
Posts Count	95
Comments Count	112
Likes Count	430
Shares Count	28
Login Frequency	10
Activity Date	18-Aug-2026
User Identifier	uid_6610
Session Id	sess_7789e

Add a comment

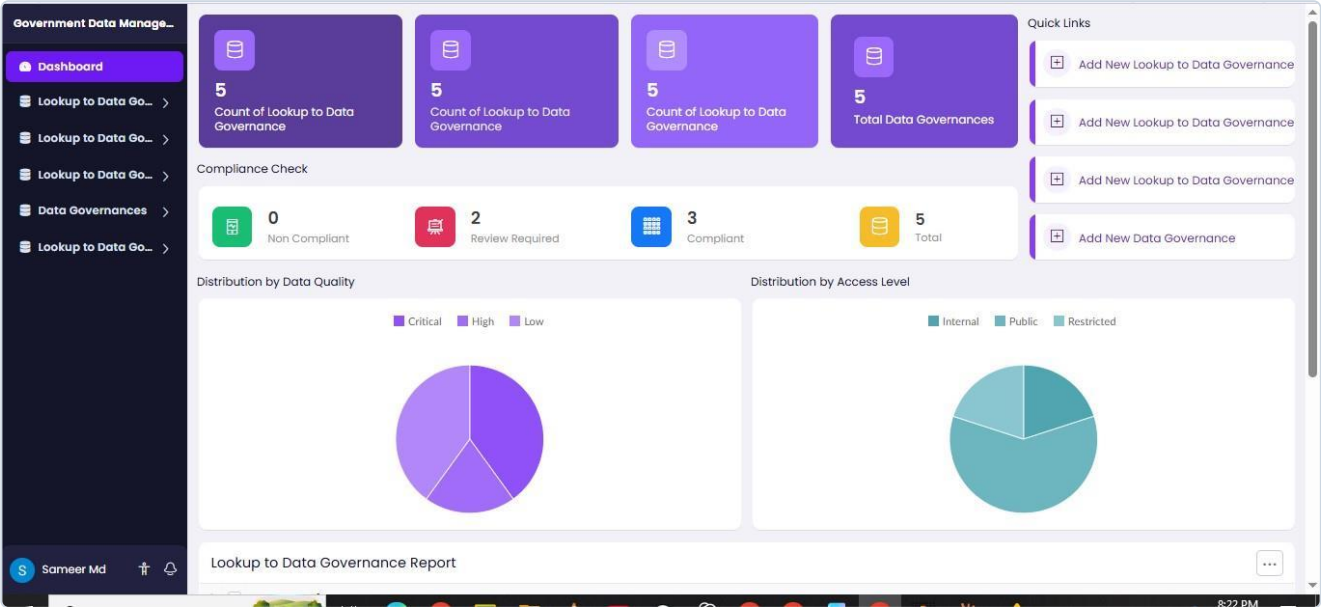
Q3 – Government Data Platform

Problem Statement: A government agency is building a national data platform combining census, economic, and geospatial datasets. Different departments need access to shared, integrated data ensuring interoperability across agencies, strict privacy, and security. Propose a governance and data management strategy.

A centralized National Data Platform provides an interoperable, unified data fabric across disparate government departments (Census, Finance, Geospatial Survey, Transport, and Health).

- **Data Ingestion & Interoperability:** Standardized REST APIs, OGC formats (GeoJSON/Shapefiles), and parquet data lakes eliminate agency silos.
- **Data Governance Framework:** Implements comprehensive metadata catalogs (e.g., Apache Atlas) defining data ownership, lineage, quality tiers, and access classification.

- Privacy & Security:** Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC) enforce least privilege. Citizen PII is pseudonymized/masked at ingestion and protected using AES-256 encryption at rest and TLS 1.3 in transit.
- Auditing:** Immutable logging tracks every inter-departmental query to ensure compliance with national data privacy statutes.



Dataset Reference	
DS-1001	DS-1003
DS-1001	DS-1002
DS-1001	DS-1003
DS-1001	DS-1002
DS-1001	DS-1003
DS-1001	DS-1004
DS-1001	DS-1005
DS-1001	DS-1002
DS-1001	DS-1003
DS-1001	DS-1002
DS-1001	DS-1003
DS-1001	DS-1004
DS-1001	DS-1005

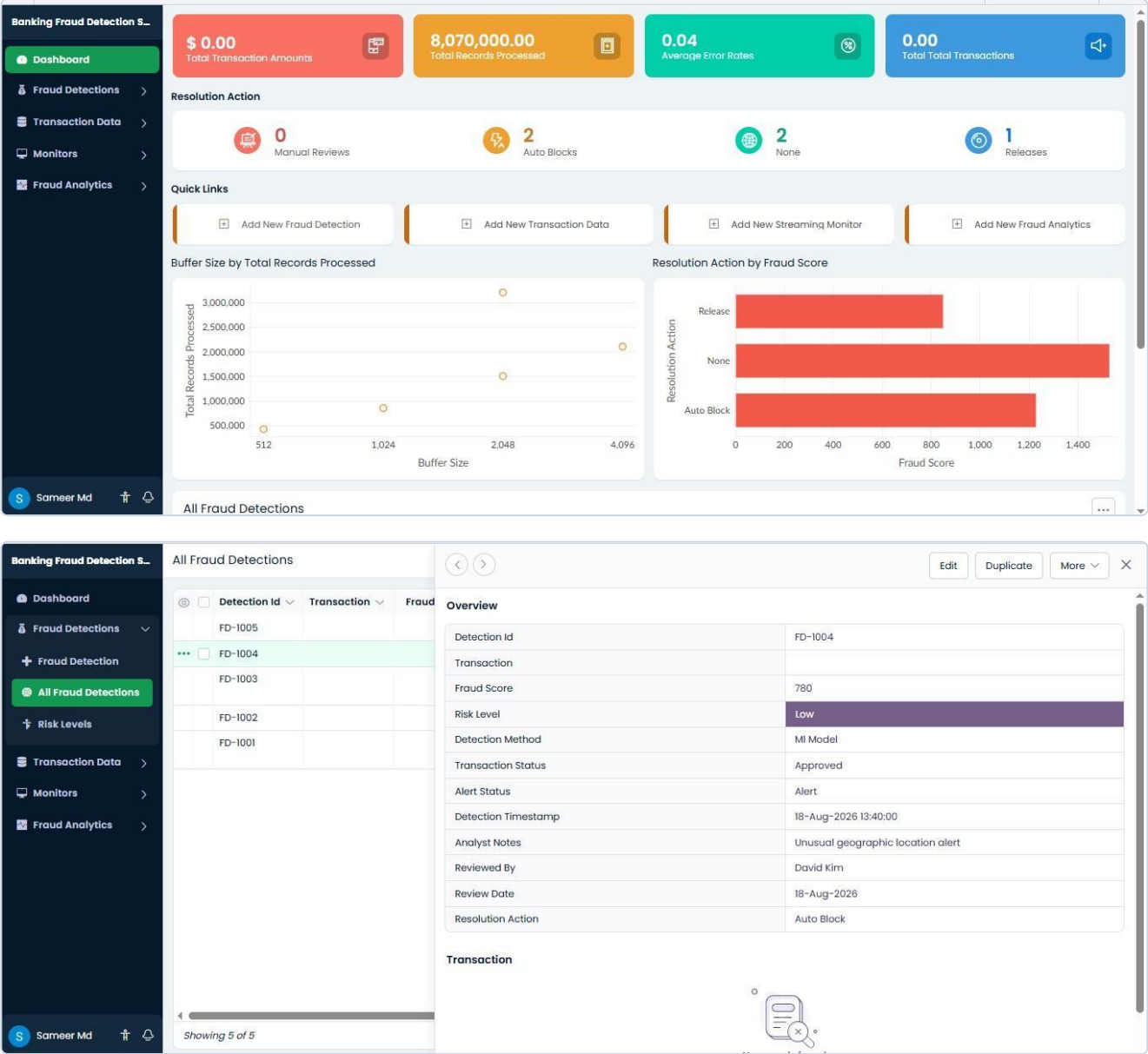
Q4 – Banking Fraud Detection

Problem Statement: A bank processes millions of financial transactions every minute and needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data with low latency and high accuracy. Design a big data pipeline and explain the technologies and algorithms implemented.

Financial fraud detection demands sub-second latency and ultra-high precision to prevent unauthorized monetary loss without causing customer friction through false positives.

- Real-Time Pipeline:** Apache Kafka ingests high-volume transaction feeds (account ID, amount, merchant terminal, geolocation, IP, device ID).

- **Low-Latency Processing: Apache Spark Streaming / Flink** performs sliding-window aggregations to identify velocity anomalies (e.g., multiple transactions in different cities within 5 minutes).
- **Machine Learning Models:** Real-time ensemble models (Isolation Forest, Gradient Boosted Decision Trees, Neural Networks) generate a real-time Fraud Score between 0 and 1000.
- **Automated Decisioning:** Low-risk transactions are approved immediately; intermediate scores trigger step-up multi-factor authentication (MFA); high scores trigger automated card blocking and fraud investigator alerts.



Q5 – Agriculture Smart Farming

Problem Statement: A smart farming system collects soil, weather, and crop data via sensors and drones to provide real-time field information. Farmers want insights to improve yield and resource usage. Design a big data solution for precision agriculture and explain how analytics supports decision-making.

Precision agriculture leverages distributed IoT field sensors (soil moisture, NPK nutrients, soil pH), automated weather stations, and drone multispectral imagery to optimize crop yield and conservation.

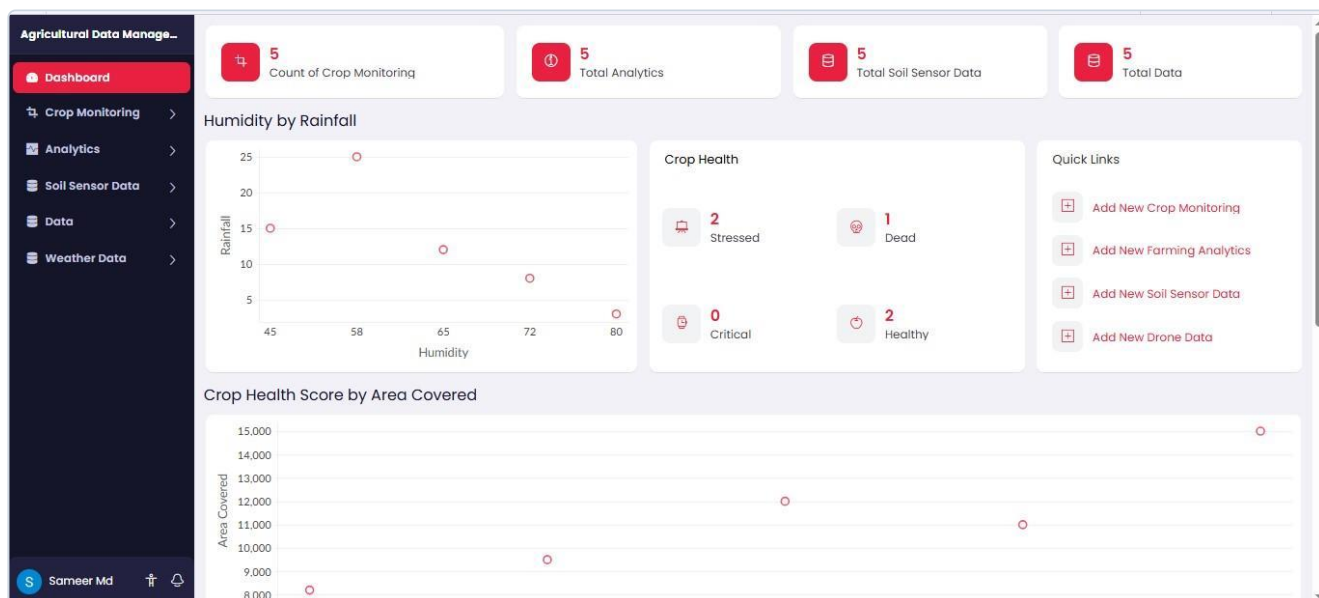
- **IoT & Drone Ingestion:** Edge gateways gather telemetry from solar-powered field sensors and upload aerial drone imagery to cloud object storage.

- **Analytics & Computer Vision:** **Apache Spark** processes spatial sensor data while CNN models analyze drone imagery (NDVI index) to locate crop disease or water stress zones.
- **Actionable Insights:** Predictive models generate optimal variable-rate irrigation and fertilization schedules, alerting farmers via mobile dashboards to apply resources only where needed.

[Soil Sensors] + [Weather Stations] + [Crop Sensors] + [Drones] → Data Collection → Cloud Storage → Big Data Processing (Spark/ML)

↓
[Soil Health Analysis] | [Yield & Disease Prediction] | [Weather Advisory]

↓
Farmer Mobile / Web Dashboard → Better Agronomic Decisions



Agricultural Data Management System - All Analytics

Analytics Id	Field Code	Irrigation
1005		Immediate
1004		Low
1003		None
1002		High
1001		High

Overview

Analytics Id	1004
Field Code	
Irrigation Recommendation	Low
Fertilizer Recommendation	Mixed
Disease Risk	560
Predicted Yield	61000
Resource Usage	1900
Recommendation Status	Pending
Soil Sensor Data	
Weather Data	

Weather Data

No records found

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User: Sameer Md